Unit 6: Introduction to Digital Graphics and Animation

Level: **2**

Unit type: **Internal**

Guided learning hours: **60**

Unit in brief

Learners will design and create digital graphics and animation in response to a client brief.

Unit introduction

Digital graphics and animation are used in nearly all modern digital products to enhance the product and meet user needs. Whether it is characters in a computer game,

the interface for a smartphone application (app), or diagrams designed to provide information on a website, digital graphics and animation make the product more engaging for the user and are vital to the success of the product.

In this unit, you will learn about the principles of graphics and animation. You will explore how graphics are used in different situations, and how the different styles, features and properties of graphics impact on their use and effectiveness.

You will learn how to design, create and edit graphics and animation to meet identified requirements, and how to work in a legal and ethical manner when producing graphics for an identified purpose.

Learning aims

In this unit you will:

A Explore the use of graphics and animation in digital products

B Design digital graphics and animation to meet a client brief

C Develop digital graphics and animation to meet a client brief.

Summary of unit

Learning aim	Key content areas	Assessment approach
A Explore the use of graphics and animation in digital products	**A1** The purpose of digital graphics and animation **A2** Properties of graphics and animation used in digital products	A report exploring the use of digital graphics and animation in a range of contexts.
B Design digital graphics and animation to meet a client brief	**B1** Generating ideas **B2** Design documents for digital graphics and animation **B3** Legal and ethical considerations	A portfolio of evidence produced in response to a brief to include: • initial design ideas • detailed design documents for graphic and animation products • a record of pre-existing assets to be used including details of copyright and other legal considerations • a commentary on ethical factors that have been considered in the designs • final digital graphic and animation files.
C Develop digital graphics and animation to meet a client brief	**C1** Digital graphics tools **C2** Digital animation tools **C3** Preparing products for use in a digital product	

Content

Learning aim A: Explore the use of graphics and animation in digital products

A1 The purpose of digital graphics and animation

Use of digital graphics to enhance digital products.

- The purpose of graphics and animation in a digital product:
  - to help explain a concept, e.g. a diagram, chart etc.
  - to improve accessibility for the user
  - to be used as an asset, e.g. button for a user interface, character in a computer game
  - to convey information, e.g. table, chart, infographic
  - to entertain, e.g. a comic, cartoon, meme.
- Characteristics of intended audiences that affect the design and use of digital graphics, e.g. age, interests, level of education.
- Use of digital design elements to help digital graphics and animations meet their intended purpose:
  - use of text – amount of text, font style, size, complexity of language, use of technical terms
  - design style – realistic/photographic, diagram, cartoon, map, 2D/3D
  - use of colour
  - image composition – foreground and background elements, use of visual hierarchy, positioning of elements, e.g. left of screen, next to specific elements.
- Use of digital effects to enhance the graphic or animation and help meet its intended purpose, e.g. blurring, rotation, fading, transitions.

A2 Properties of graphics and animation used in digital products

The properties of digital graphics and animation and the impact on their use and appropriateness for inclusion in digital products.

- Difference in the storage and rendering of bitmap/raster-based and vector-based graphics.
- Typical uses of raster-based and vector-based graphics or animation:
  - raster-based, e.g. photographs, website graphics, print advertising, packaging, computer game assets
  - vector-based, e.g. user interfaces, logos, illustrations, buttons for interactive media products, icons, fonts.

- Properties of graphics and animation that impact on their use and effectiveness – resolution, dimensions, file type, colour mode, compression, file size.
- Impact of the properties of graphics and animations on the:
    - effectiveness of the product, e.g. clarity of message, engagement, readability, ease of use
    - appropriateness for the intended audience
    - load times
    - file size
    - compatibility.

Learning aim B: Design digital graphics and animation to meet a client brief

Conveying design ideas for digital graphics and animations for technical and non-technical audiences in response to a brief.

B1 Generating ideas

- Methods for generating initial ideas in response to a brief:
  - research of similar products
  - mood boards
  - brainstorming
  - thumbnail sketches.
- Seeking feedback from others and refining ideas.

B2 Design documents for digital graphics and animation

- Project design brief to include details of:
  - the target product
  - the target audience
  - any technical considerations, e.g. devices it will be used on, installed application or web-based, performance requirements.
- Design documentation for digital graphics including:
  - a detailed sketch of each graphic
  - colour palettes
  - details of text to be used – content, font/typeface, sizes, styles
  - how digital effects will be used
  - properties of the target graphics, e.g. file size, file types, compression.
- Design documentation and storyboards for digital animation including:
  - summary of the overall story or message
  - detailed sketches of key frames/scenes
  - colour palettes
  - details of text to be used – content, font/typeface, sizes, styles
  - use of digital effects and sounds
  - timings and frame rates
  - intended properties of the target animation, e.g. file size, file types, compression.

- Use of assets/content created by others:
    - identify assets/content to be used
    - summary of any editing or manipulation of the asset that will be required,

e.g. cropping, change of file type.

B3 Legal and ethical considerations

Legal and ethical considerations when planning, preparing and developing digital graphics and animation.

- Recording and attributing sources of assets/content created by others:
    - the specific source of the asset, e.g. URL
    - licensing or copyright restrictions to be considered
    - evidence of permissions for use (if required).
- Ethical factors to consider when producing digital content:
    - privacy and confidentiality
    - representation of people of different genders, races, religious beliefs, etc.
    - inappropriate or age-restricted content
    - accessibility.

Learning aim C: Develop digital graphics and animation to meet a client brief

C1 Digital graphics tools

Software tools to produce and prepare raster-based and vector-based graphics to meet the requirements of a brief.

- Tools and techniques for creating digital graphics:
  - freehand draw
  - grouping
  - colour balance
  - filters
  - selection
  - hue and saturation
  - masking
  - layering
  - retouching
  - opacity/transparency
  - importing from external sources
  - editing and combining paths.

C2 Digital animation tools

Software tools to produce digital animation to meet the requirements of a brief.

- Producing and preparing vector-based and raster-based assets for inclusion in an animation as required.
- Tools and techniques for creating digital animation:
  - frame rates
  - onion skinning
  - tweening
  - transitions
  - camera angles
  - movement
  - picture duration
  - rendering

- importing and applying sound and other externally created assets.

C3 Preparing products for use in a digital product

Preparing digital graphics and animation so that they are suitable for use in an identified digital product.

- Saving, exporting and optimising final graphics and animation so that they are ready for use, including appropriate selection of:
  - resolution
  - dimensions
  - bit depth
  - colour modes
  - file type
  - compression methods
  - file sizes.

Assessment criteria

Pass	Merit	Distinction
Learning aim A: Explore the use of graphics and animation in digital products		
A.P1 Describe the use of digital graphics and animations in identified digital products. **A.P2** Describe how the design and properties of digital graphics and animations impact on the effectiveness of identified products.	**A.M1** Assess the effectiveness of digital graphics and animation in identified products.	**A.D1** Evaluate the effectiveness of the use of digital graphics and animation in identified products.
Learning aim B: Design digital graphics and animation to meet a client brief		**B.D2** Produce comprehensive designs for digital graphics and animation, including an assessment of relevant legal and ethical factors, in response to a client's brief.
B.P3 Produce basic designs for digital graphics and animation to meet the client's brief. **B.P4** Identify legal and ethical factors considered when responding to the client's brief.	**B.M2** Produce detailed and effective designs for digital graphics and animation, including an explanation of relevant legal and ethical factors, in response to a client's brief.	
Learning aim C: Develop digital graphics and animation to meet a client brief		**C.D3** Produce effective and optimised digital graphics and animation making use of a range of software tools to fully meet the identified client's needs.
C.P5 Produce digital graphics and animation that make basic use of some appropriate software tools in response to a client's brief.	**C.M3** Produce digital graphics and animation making use of a range of software tools to meet the identified client's needs.	

Essential information for assignments

The recommended structure of assessment is shown in the unit summary, along with suitable forms of evidence. *Section 6 Internal assessment* gives information on setting assignments and there is also further information on our website.

There is a suggested maximum number of three summative assignments for this unit. The relationship of the learning aims and criteria is:

Learning aim: A (A.P1, A.P2 A.M1, A.D1)

Learning aims: B and C (B.P3, B.P4, C.P5, B.M2, C.M3, B.D2, C.D3)

Further information for teachers and assessors

Resource requirements

For this unit, learners must have access to hardware and software that allow them to plan and create digital graphics and animation. Software resources may include:

- Inkscape – Open source vector graphics creation and editing software – https://inkscape.org/
- GIMP – Open source graphic editor – https://www.gimp.org/
- Blender – Open source 3D creation suite – https://www.blender.org/
- Pencil – Open source 2D animation tool – https://www.pencil2d.org/
- Synfig Studio – Open source 2D animation software – https://www.synfig.org/

Essential information for assessment decisions Learning aim A

While there is no requirement for a specific number of products, it is recommended

that learners examine at least three different digital products. Learners may select the products themselves or tutors may direct learners to suitable products. The products selected must provide suitable opportunity for learners to consider a range of intended audiences and purposes and different uses of digital graphics and animation. Learners should present their evaluation in the form of a report.

For Distinction standard, learners must evaluate the effectiveness of graphics and animation in a number of different digital products.

The evaluation must consider how different elements that make up the identified digital graphics and animations help them meet their intended purpose (including use of text, style, colour, composition and digital effects). Learners should also consider how the properties of the graphics and animations impact on the way they are used, and reach a value judgement as to how well the graphics meet the requirements of the audience and product. Learner must provide supported justifications for the assessment they make.

At this level learners should make appropriate use of technical vocabulary to effectively support the points they make in their report.

For Merit standard, learners must assess the effectiveness of graphics and animation in a number of different digital products.

Their assessment should consider how different elements that make up the identified digital graphics and animations help them meet their intended purpose (including use of text, style, colour, composition and digital effects), but at this level some of the points they make may be more generic and the judgment may not always be supported.

Learners must explore the properties of the graphics and animation and describe how these may impact on their use, but there may be some minor technical inaccuracies, or the exploration of the technical aspects will, at times, not go into sufficient depth.

They will consider how this will impact on the product they are included in, but consideration of subsequent impact on the user will not be fully explored.

At this level learners should make mostly appropriate use of technical vocabulary to effectively support the points they make in their report.

For Pass standard, learners will describe the use of digital graphics and animations in identified digital products. They will include a description of how the images are designed and constructed and how these relate to the intended audience.

Learners will describe how the design and properties of digital graphics and animations impact on the effectiveness of identified products. At this level learner responses will make some accurate observations but will not demonstrate multiple chains of reasoning. For example, they might identify that the size of an image could cause problems with load times, but they are unlikely to expand on this.

At this level learners will make some appropriate use of technical vocabulary.

Learning aim B

For Distinction standard, learners must produce comprehensive designs for digital graphics and animation in response to a given scenario.

Learners' research into similar products could be in the form of annotated screen shots of existing products. Their annotations will show insightful consideration of the features of these products and how these can be applied to their own designs. Learners' draft ideas in response to the brief should demonstrate a wide range of design-based skills, for example using mood boards, records of brainstorming sessions and thumbnail sketches. At this stage learners do not need to produce detailed designs for every graphic and animation to be created, but should provide enough detail to allow

others to provide feedback on the merits of the suggested ideas, and potential improvements/refinements.

Learners should use the feedback that they receive, along with their initial ideas, to develop a set of detailed design documentation for the graphics and animation they intend to create. The design documentation should be of sufficient clarity and detail (as listed in content area B2) so that, if required, a third party could create the intended products using the learner's design documentation, with minimal difficulty.

Learners must provide a detailed record of the sources of any assets created by others,

e.g. sound files, graphics that they intend to use in their products. Their sources table should provide accurate details of where the original source can be found and any editing that will be required. Learners should also, either as a separate document or as part of their sources table, provide relevant details of any restrictions and permissions, as listed in B3. Learners should also produce a supporting document providing an accurate and detailed description of how they have considered ethical factors in the design decision that they have made.

For Merit standard learners must produce detailed and effective designs for digital graphics and animation in response to a given scenario.

Learners' research into similar products that exist could be in the form of annotated screen shots of existing products. The annotations will show some a considered assessment of the features of these products and how these can be applied to

their own designs.

Learners' draft ideas in response to the brief will demonstrate a range of design-based skills, for example using mood boards. Learners' ideas as to what will be produced will be appropriate and generally well-considered, but they may not always be sufficient for others to make detailed contributions as to the merits of the suggested ideas, and potential improvements/refinements.

At this level there will be some evidence as to how the feedback that they received on their initial ideas influenced designs, but this will not always be clear.

Learners will produce a set of detailed design documentation for the graphics and animation they intend to create. At this level the design documentation should be of reasonable quality, so that a third party could mostly create the intended products, but there may be some minor difficulties due to lack of detail or clarity in places.

Learners must record the sources of any assets created by others, e.g. sound files, graphics that they intend to use in their products. Their sources table should provide details of where the original source can be found, and any editing that will be required but this may not always be clear. Learners should also provide details of any restrictions and permissions that are relevant (as listed in B3.1), either as a separate document or as part of their sources table. Learners should also produce a supporting document describing how they have considered ethical factors in the design decisions that they have made, but this may not always be fully supported or clearly related to their product. Their documents will be mostly well-reasoned but there may be some inaccuracies or misconceptions.

For Pass standard, learners must produce basic designs for digital graphics and animation in response to a given scenario.

Learners' research into similar products that exist could be in the form of annotated screen shots of existing products. The annotations will show that they have taken into consideration some of the relevant features of similar products. Descriptions as to how these can be applied to their own designs are likely to be appropriate, but may be superficial or generic.

Learners' draft ideas in response to the brief must demonstrate an understanding of design processes, for example using mood boards. Their ideas will give some indication as to what will be produced, however these may not always be clear, making it difficult for others to make contributions as to the merits of the suggested ideas, or any potential improvements/refinements.

At this level there may be some evidence of appropriate use of feedback received to improve their work, but it will be limited.

Learners will produce a set of basic design documentation for the graphics and animation they intend to create. The product and some key design features/tools will be identified in the design documentation, however, it is unlikely that a third party would be able to use them to create the intended products without significant assistance.

Learners must record the sources of any assets created by others, e.g. sound files, graphics that they intend to use in their products. Their sources table should provide mostly accurate details of where the original source can be found. But details of editing required is likely to be omitted or very weak.

Learners will provide some appropriate details of any legal and ethical factors they have considered. However, at this level it is likely to be quite superficial and/or generic.

Learning aim C

Learners will produce digital graphics and animation in response to a brief, as detailed in plans.

For Distinction standard, learners must make effective use of tools and features provided in graphic and animation software to produce a range of graphics and

animations to meet the identified client needs. The products produced will show clear brand identity and be suitable for the target audience. At this level, the work produced will show a good understanding of technical aspects of digital graphics and animation, as well as good aesthetic considerations to produce highly effective products.

The final versions of the products will be optimised for inclusion in the target product.

The learners' final digital graphics and animation should be exported into file formats that are suitable for inclusion in the target product, but there is no need to produce the target product for this unit's assessment.

For Merit standard, learners must make mostly appropriate use of tools and features provided in graphic and animation software to produce a range of graphics and animations to mostly meet the identified client needs.

The products produced will show clear brand identity and be suitable for the target audience. At this level, the work produced will show a sound understanding of technical aspects of digital graphics and animation, as well as sound aesthetic considerations to produce highly effective products. However, while most are appropriate, there may still be some features chosen that are less effective or less appropriate for the intended purpose.

The final versions of the products will be exported into file formats that are suitable for inclusion in the target product but may not be fully optimised.

For Pass standard, learners must make some appropriate use of tools and features provided in graphic and animation software to produce a range of basic graphics and animations that make some appropriate use of a limited range of digital graphics and animation tools to meet some of the identified client needs.

The products produced will be mostly suitable for the target audience and/or product but will lack a cohesive brand identity.

At this level, the work produced will show a basic understanding of technical aspects of digital graphics and animation, but there may be some tools or features that are not applied correctly.

The graphics will show some understanding of basic aesthetic considerations, for example they may have chosen and applied suitable colour schemes, but may not always consider relative sizing of components as effectively, or may not apply more advanced tools which would make the work fit the product more cohesively.

The graphics and animations the learner produces will meet most aspects of the brief at a basic level.

The final versions of the products may be exported but these may not always be suitable for inclusion in the target product.

Links to other units and curriculum subjects

This unit links to:

- Unit 1: Using IT to Support Information and Communication in Organisations
- Unit 7: Introduction to Website Development
- Unit 8: Introduction to App Development

- Unit 9: Introduction to Game Design
- International GCSE/core curriculum in Information Technology.

Employer involvement

This unit would benefit from employer involvement in the form of:

- guest speakers and interview opportunities
- work experience
- business material as exemplars
- visits to appropriate business organisations.

Opportunities to develop transferable employability skills

In completing this unit, learners will have the opportunity to develop research and planning skills as well as creative skills.

They also have opportunities to develop communication skills and how to combine these with creative skills to communicate in different ways depending on purpose and audience.